

Bus-Level Electricity Load Forecasting Using Machine Learning

Machine Learning Forecasting Assignment

1. Introduction

The objective of this project was to develop a machine learning forecasting pipeline for hourly electricity demand at the bus level. The assignment focused on predicting future electricity load using historical load observations while avoiding data leakage and preserving operational forecasting realism.

Two forecasting horizons were evaluated:

1. next-day forecasting
2. next-month forecasting

Historical PJM ISO bus-level load data from 2022 through 2025 was used to train and evaluate the forecasting models. The forecasting workflow emphasized feature engineering, lag-based temporal modeling, rolling forecast generation, and comparative evaluation against baseline methods.

2. Forecasting Strategies

Two forecasting strategies were considered in this project.

The primary implemented strategy was direct bus forecasting. Under this approach, a machine learning model directly predicts the load for each individual bus using historical load patterns and engineered temporal features.

A second conceptual strategy involved forecasting total zone-level demand first and then distributing predicted zone demand to individual buses according to historical bus-level demand shares. This approach may provide more stable long-horizon forecasts because aggregated zone demand is generally smoother and less noisy than individual bus demand. However, this method may also reduce sensitivity to localized bus-level fluctuations.

The final implementation focused primarily on the direct bus forecasting strategy.

3. Baseline Forecasting Methods

A baseline forecasting method was implemented to provide a benchmark against which the machine learning model could be compared.

The primary baseline used the load from the same hour during the previous week:

$$\hat{y}(t) = y(t-168)$$

This baseline is commonly used in electricity forecasting because power demand often exhibits strong weekly seasonality patterns.

An additional conceptual baseline involved calculating historical average demand grouped by

bus, hour, and weekday. Although this approach was not fully implemented, it represents another standard forecasting benchmark.

4. Machine Learning Model

The primary forecasting model used in this project was LightGBM Regressor. LightGBM was selected because it performs efficiently on large tabular datasets and can capture nonlinear relationships between temporal features and future electricity demand.

The model configuration included:

- 100 boosting estimators
- learning rate of 0.05
- maximum tree depth of 8

LightGBM provided a computationally efficient forecasting framework while maintaining strong predictive performance on structured time-series features.

5. Feature Engineering

Several temporal and historical features were engineered for forecasting.

Temporal features included:

- hour of day
- day of week
- month
- weekend indicator

Historical lag features included:

- 24-hour lag demand
- 168-hour lag demand (same hour previous week)

Rolling statistical features were also generated, including a rolling 24-hour mean demand.

These features allowed the model to capture:

- daily load cycles
- weekly seasonality
- recent demand trends
- temporal autocorrelation structures

Lag-based features were among the most influential predictors because electricity demand is highly dependent on recent historical behavior.

6. Prevention of Data Leakage

Avoiding future information leakage was a critical component of the forecasting workflow.

Several precautions were implemented:

- lag variables only used historical observations
- rolling statistics were shifted before aggregation
- chronological train/test splitting was used

- no future target values were included in feature generation

Rolling features were implemented using shifted historical observations to ensure that information from the current target period was not accidentally included in predictor variables.

These precautions ensured that forecasts only used information that would realistically have been available at forecast creation time.

7. Train-Test Structure

The dataset was divided chronologically in order to preserve realistic forecasting conditions.

The training period included data from:
2022-01-01 through 2024-12-31

The testing period included data from:
2025-01-01 through 2025-12-31

This structure simulated operational forecasting, where historical information is used to predict future unseen demand.

8. Next-Day Forecasting Pipeline

The next-day forecasting pipeline generated rolling daily forecasts throughout the 2025 test period.

For each forecast date:

1. historical features were generated
2. the next 24 hours were selected as prediction targets
3. the trained model produced hourly bus-level forecasts

This approach simulated operational short-term forecasting conditions commonly used in electricity system management.

9. Next-Month Forecasting Pipeline

The next-month forecasting pipeline generated rolling monthly forecasts.

For each monthly forecast creation date:

1. the subsequent month was selected as the forecasting horizon
2. hourly bus-level forecasts were generated for the entire month
3. forecast outputs were stored in the required submission format

Compared with next-day forecasting, next-month forecasting is significantly more difficult because uncertainty accumulates over longer forecasting horizons.

10. Forecast Evaluation

Forecasting performance was evaluated using three standard metrics:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)

- Weighted Mean Absolute Percentage Error (WMAPE)

These metrics were evaluated separately for:

- next-day forecasting
- next-month forecasting
- bus-level forecasting accuracy
- zone-level aggregated forecasting accuracy

The machine learning model generally outperformed the baseline forecasting approach, particularly during stable demand periods with strong recurring temporal structure.

11. Model Performance Discussion

The forecasting model performed well during:

- regular weekday demand cycles
- stable seasonal demand periods
- recurring temporal demand structures

The model performed less effectively during:

- anomalous demand spikes
- irregular holiday behavior
- abrupt load changes
- long-horizon monthly forecasting

One important limitation was the absence of weather-related features. Electricity demand is strongly influenced by temperature and extreme weather conditions, and the model could not fully capture these effects without meteorological information.

12. Limitations and Future Improvements

Several limitations remain in the current implementation.

First, the next-month forecasting pipeline did not implement fully recursive autoregressive forecasting. Future lag variables were still partially dependent on observed historical values rather than recursively generated forecasts.

Second, the forecasting pipeline did not include:

- weather variables
- holiday calendars
- economic indicators
- probabilistic uncertainty intervals

Potential future improvements include:

- recursive forecasting implementation
- weather forecast integration
- walk-forward validation
- comparison with XGBoost and neural forecasting models
- probabilistic forecasting methods
- complete zone-level forecasting pipelines

These additions would likely improve forecasting realism and long-horizon predictive accuracy.

13. Conclusion

This project developed a complete machine learning forecasting workflow for hourly electricity demand prediction at the bus level. The forecasting pipeline incorporated lag-based temporal features, rolling forecasting procedures, leakage prevention methods, and comparative baseline evaluation.

The LightGBM forecasting model demonstrated strong predictive performance for short-term electricity forecasting tasks and provided a scalable framework for future forecasting improvements. The project also highlighted the increasing difficulty of long-horizon forecasting and the importance of careful temporal validation and feature engineering in operational forecasting systems.